

Linear Algebra Practice Set

INF1004: Mathematics 2

Part I: Matrix Operations

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ 2 & 1 \end{pmatrix}$, and $C = \begin{pmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \end{pmatrix}$.

1. **Basic Matrix Arithmetic:** Let $M = A + B$, $N = 3A - 2B$, and $P = AB$.

- (a) What is the sum of the elements in the first row of matrix M ?
- (b) What is the value of the element in the second row, first column of N ?
- (c) What is the trace (sum of the main diagonal elements) of matrix P ?
- (d) Calculate the determinant of the matrix product BA .
- (e) Calculate the sum of all elements in the matrix $(A^T + B^T)$.

2. **Matrix Inverses and Equations:**

- (a) What is the value of the determinant of A ?
- (b) Let A^{-1} be the inverse of A . What is the value of the element in the first row, second column of A^{-1} ?
- (c) Let X be the 2×2 matrix that satisfies the equation $2X + B = A$. What is the trace of X ?
- (d) Let $Y = AC$. What is the value of the element in the second row, third column of Y ?
- (e) A store sells 2 apples and 1 banana for \$5, and 1 apple and 2 bananas for \$4. Let a be the unit price of an apple. What is the value of a ?

Part II: Vectors, Lines, and Planes

Let $\vec{u} = (1, 0, 2)^T$, $\vec{v} = (-1, 2, 1)^T$, and $\vec{w} = (0, 1, -1)^T$ in $\mathbb{R}^3$.

3. **Vector Products and Angles:**

- (a) Calculate the dot product $\vec{u} \cdot \vec{v}$.
- (b) Let $\vec{c} = \vec{u} \times \vec{v}$. What is the y -component of $\vec{c}$?
- (c) What is the square of the magnitude of the vector $\vec{u} \times \vec{v}$?
- (d) Let θ be the angle between vectors $\vec{u}$ and $\vec{v}$. What is the exact value of $30 \cos^2(\theta)$?
- (e) Calculate the dot product $\vec{u} \cdot \vec{w}$.

4. **Geometry in $\mathbb{R}^3$:**

- (a) A line passes through the point $(1, 2, 3)$ and is parallel to $\vec{u}$. If the point $(x, y, 7)$ lies on this line, what is the value of x ?
- (b) A plane passes through the origin and has $\vec{u}$ as its normal vector. If the point $(A, 3, -1)$ lies on this plane, what is the value of A ?

- (c) A plane contains the points $P(1, 0, 0)$, $Q(0, 1, 0)$, and $R(0, 0, 1)$. What is the z -intercept of this plane?
- (d) The line defined by $\vec{r} = (1, 1, 1)^T + t(1, 0, 2)^T$ intersects the plane $x + y + z = 5$ at a specific point (p_x, p_y, p_z) . What is the value of $3p_x$?
- (e) Let d be the shortest distance from the origin to the plane $x + y + z = 1$. What is the value of $d \times \sqrt{3}$?

Part III: Systems of Linear Equations

5. Solving Methods and Determinants:

- (a) Use Cramer's rule to solve the system: $2x + y = 5$ and $x - y = 1$. What is the value of $x + y$?
- (b) Solve the system: $x + y + z = 6$, $2x - y + z = 3$, and $x + 2y - z = 2$. What is the value of the product $x \times y \times z$?
- (c) The system of equations $3x - 2y = 4$ and $x + y = 5$ can be represented as $A\vec{x} = \vec{b}$. What is the determinant of the coefficient matrix A ?
- (d) Let A^{-1} be the inverse of the coefficient matrix from the previous part (5c). What is the sum of all elements in the matrix $5A^{-1}$?
- (e) Consider the system: $x + 2y = 4$ and $2x + 4y = 8$. If $y = 0$, what is the corresponding value of x for a valid solution?

6. Applications and 3x3 Matrices:

- (a) Consider the system: $x + y = 2$ and $x + y = 3$. How many solutions exist for this system?
- (b) Let $M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$. What is the determinant of M ?
- (c) Let M^{-1} be the inverse of M . What is the sum of the diagonal elements of the matrix $2M^{-1}$?
- (d) Solve the system using Gaussian Elimination: $x + y = 2$, $y + z = 2$, and $x + z = 2$. What is the value of $x^2 + y^2 + z^2$?
- (e) A factory produces items A and B. Item A requires 2 hours of labor, and item B requires 3 hours. Total labor available is 12 hours. Additionally, A and B each require 1 unit of material, with 5 units of material available. Solve using Gaussian Elimination. How many of item A are produced?

Part IV: Eigenvalues, Eigenvectors, and Diagonalization

7. 2x2 Eigenvalue Analysis:

- (a) Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. What is the sum of the eigenvalues of A ?
- (b) What is the product of the eigenvalues of matrix A from part (7a)?
- (c) For the larger eigenvalue of A , let its corresponding eigenvector be written in the form $(1, y)^T$. What is the value of y ?
- (d) Let $B = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$. What is the absolute difference between the eigenvalues of B ?

- (e) Let $C = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$. What is the smallest (most negative) eigenvalue of C ?

8. 3x3 Matrices and Diagonalization:

- (a) Verify that $\vec{v} = (1, 1)^T$ is an eigenvector of $D = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$. What is its corresponding eigenvalue?
- (b) What is the other eigenvalue of matrix D from part (8a)?
- (c) Let $M = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$. What is the sum of the roots of its characteristic polynomial?
- (d) What is the largest eigenvalue of matrix M from part (8c)?
- (e) The matrix A from part (7a) can be diagonalized as PDP^{-1} where D is a diagonal matrix of eigenvalues. What is the sum of all elements in D ?

Part V: Orthonormality and Singular Value Decomposition

9. Orthogonality and Basic SVD Properties:

- (a) The matrix $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is orthogonal. What is the determinant of Q ?
- (b) Let $R = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$ and $I = R^T R$. What is the sum of all elements in the matrix I ?
- (c) For $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$, let $K = A^T A$. What is the value of the element $K_{2,2}$?
- (d) What is the largest singular value (σ_1) of the matrix A from part (9c)?
- (e) What is the sum of all the singular values of matrix A ?

10. Advanced SVD Computation: Let $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$.

- (a) Let $N = B^T B$. What is the trace of matrix N ?
- (b) The singular values of B are σ_1 and σ_2 . What is the exact value of $\sigma_1^2 + \sigma_2^2$?
- (c) Determine the rank (defined as the maximum number of linearly independent rows or columns) of matrix B .
- (d) Let $P = CC^T$. What is the product of the main diagonal elements of matrix P ?
- (e) Let σ_1 and σ_2 be the non-zero singular values of C . What is the exact value of $\sigma_1^2 \times \sigma_2^2$?

Answers

Part I

1. Basic Matrix Arithmetic

- (a) 2
- (b) 5
- (c) 5
- (d) -4
- (e) 12

2. Matrix Inverses and Equations

- (a) -2
- (b) 1
- (c) 2
- (d) 11
- (e) 2

Part II

3. Vector Products and Angles

- (a) 1
- (b) -3
- (c) 29
- (d) 1
- (e) -2

4. Geometry in $\mathbb{R}^3$

- (a) 3
- (b) 2
- (c) 1
- (d) 5
- (e) The distance has to be magnitude of a vector perpendicular to the plane. So, it has to be a scaled version of the normal vector to the plane, which is $(1, 1, 1)^T$, from the coefficients of the plane equation. So, the distance vector is $(\lambda, \lambda, \lambda)^T$. However, distance vector intersects the plane, so $\lambda + \lambda + \lambda = 1$, or $\lambda = \frac{1}{3}$. Distance vector is $(1/3, 1/3, 1/3)$ which has a magnitude of $1/\sqrt{3}$, which is d . So, the required value of $d \times \sqrt{3} = 1$.

Part III

5. Solving Methods and Determinants

- (a) 3
- (b) 6
- (c) 5
- (d) 5

(e) 4

6. Applications and 3x3 Matrices

(a) 0

(b) 2

(c) 3

(d) 3

(e) 3

Part IV

7. 2x2 Eigenvalue Analysis

(a) 4

(b) 3

(c) 1

(d) 4

(e) -2

8. 3x3 Matrices and Diagonalization

(a) 6

(b) 2

(c) 8

(d) 4

(e) 4

Part V

9. Orthogonality and Basic SVD Properties

(a) 1

(b) 2

(c) 4

(d) 3

(e) 5

10. Advanced SVD Computation

(a) 3

(b) 3

(c) 2

(d) 2

(e) 2